

FADIL KARIM

COMPUTATIONAL BIM ENGINEER



SUMMARY

Expert in Revit, AutoCAD, and Bentley Suites. Specialist in Computational Design with Grasshopper, Dynamo, and Python. Experienced in building AI agents for Revit. Proficient Python coder with a deep understanding of data management & structures. Skilled in implementing AI into BIM workflows, using AI to make live changes to and analyse Revit models. Creator of a YouTube channel with 7k+ subscribers where I teach BIM modelling with Revit.

CHANNELS



Ottoman AI YouTube Channel
Revit, Grasshopper & AI Tutorials
7k+ Subscribers
[YouTube.com/@OttomanAI](https://www.youtube.com/@OttomanAI)

SOFTWARE SKILLS

3D BIM



COMPUTATIONAL DESIGN AND CODING



GRAPHIC DESIGN



EDUCATION

- 2021 - 2023
 - King's College London
 - Applied Neuroscience, PGCert
- 2010 - 2014
 - University of Birmingham
 - Civil Engineering, BEng
- 2008-2010
 - St. Joseph's College
 - A-Levels: Maths (A), Further Maths (B), Physics (C)

EXPERIENCE SUMMARY

SECTOR EXPERIENCE

Building Structures | 4 yr 11 mo
Water Infrastructure | 4 yr 1 mo
MEP | 1 yr 8 mo
Rail | 1 yr 2 mo
Temporary Structures | 8 mo

ROLE EXPERIENCE

BIM Manager | 1 yr 3 mo
AI Engineer | 6 mo
Senior BIM Technician | 8 yr 2 mo
BIM Technician | 3 yr 1 mo

TECHNICAL EXPERIENCE

BIM for Construction

3D Modelling - Revit, MicroStation/AECOSim, Civil 3D, AutoCAD

- In-depth knowledge of BIM data.
- 13+ years experience building 3D models of Structures (Buildings, Rail), MEP (Lighting, Drainage pipes, Electrical ducts), and Architecture (Rail).
- 7+ years experience building Parametric Revit Families.
- 13+ years experience creating drawings for Structures, MEP & Architecture.
- 5+ years experience in developing automation for BIM Modelling.

BIM Management

- Managed and coordinated BIM Technicians and BIM Coordinators.
- Developed BIM Execution Plans for MEP Lighting and Power Cables.
- Developed and upheld project standards, workflows, and information management procedures in accordance with BS EN ISO 19650 and UK BIM standards.
- Led BIM coordination activities including clash detection, model federation, design reviews, and issue resolution across disciplines.
- Oversaw the management and exchange of project data between multiple software platforms and Common Data Environments (CDE).
- Liaised with clients, consultants, and stakeholders to ensure alignment of BIM deliverables with project objectives.

DATA ENGINEERING for BIM

Automation - Python, C#, Dynamo & Grasshopper

- Deep understanding of Data & Information Management.
- Expert Python, Dynamo and Grasshopper.
- Able to integrate all CAD/data file types into a single model.

I developed the following automations to cut days of work down to minutes. All were shared with other team members used them without the need for expertise or additional software licenses.

Structures

- Extend/Align bottom structural elements to topography.
- Create & place hundreds of framing for complex structures.
- Place & size columns/beams directly from engineer's raw excel files.
- Create hundreds of named and numbered sheets in seconds.
- AI Agents for Revit API for 3D modelling via text prompt.
- AI Agents for spacial analysis of CAD (.dwg) files via text prompt.
- Automated optimised warehouse design in Grasshopper & Dynamo.

MEP

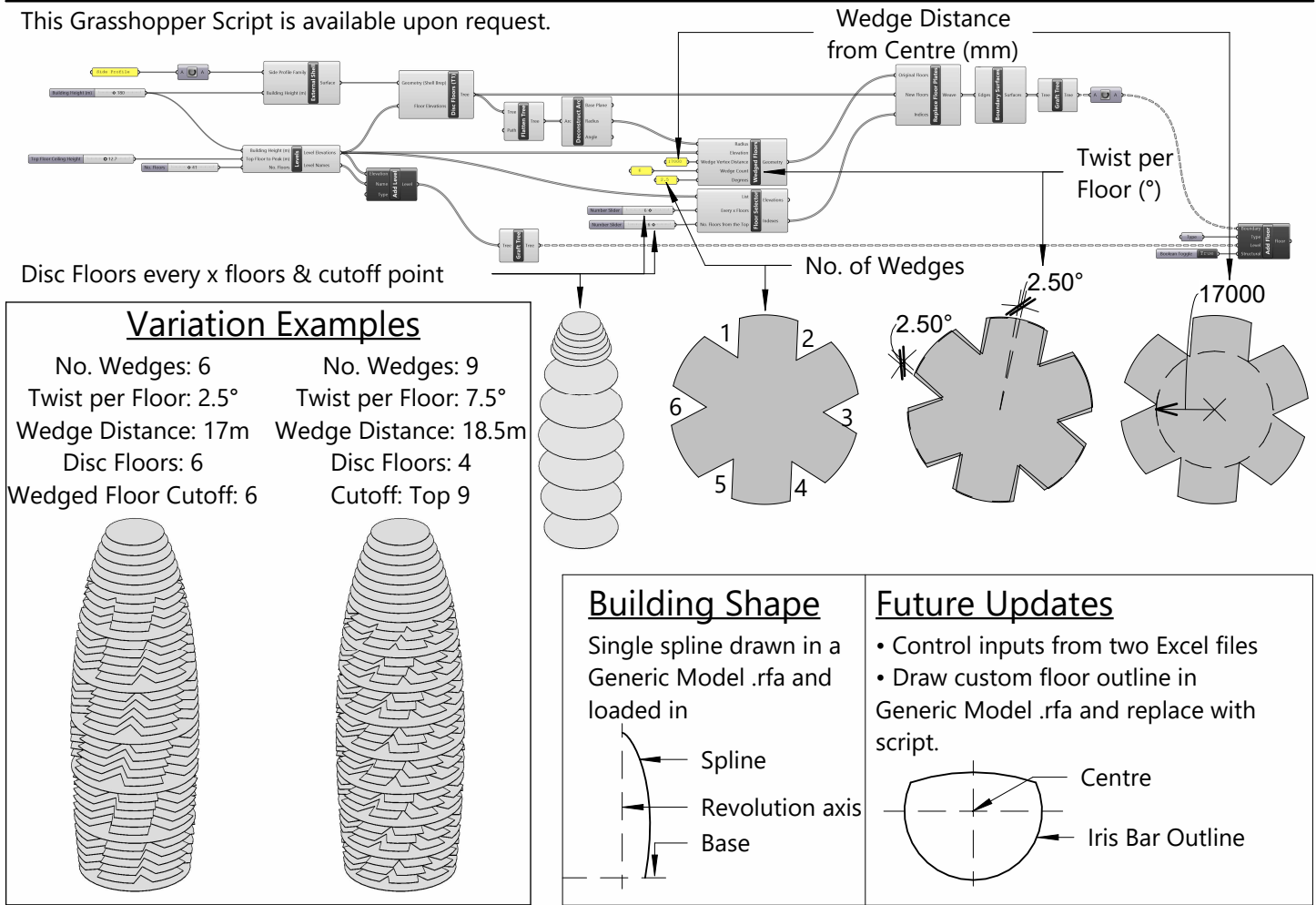
- Place and size fence & column elements on 2D model lines.
- Create concrete encasements for thousands of cable ducts runs.
- Place 3D manholes and pipes in Revit from client's .dwg files.
- Populate tens of thousands of COBie parameters in multiple models.
- Place thousands of lights directly from CAD (.dwg) files.
- Annotate hundreds of pipe slopes & diameters with custom text formats in a single click with pyRevit custom buttons.

COMPANY HISTORY

Glent Group | Senior BIM Technician | Contract | Mar 2026 – Present
Amazon | AI Engineer | Contract | Oct 2025 – Mar 2026
EGIS Group | BIM Manager | Contract | Dec 2024 – Oct 2025
Pell Frischmann | Senior BIM Technician | Contract | Sep 2024 – Dec 2024
Whitfield Consulting | BIM Manager | Contract | May 2024 – Aug 2024
ES Global | Senior BIM Technician | Contract | Jan 2024 – May 2024
Eckersley O'Callaghan | Senior BIM Technician | Contract | Oct 2023 – Jan 2024
Buro Happold | Senior BIM Technician | Contract | Nov 2022 – Sep 2023
GHD | Senior BIM Technician | Contract | Jan 2022 – Nov 2022
Robert Bird Group | Senior BIM Technician | Contract | Aug 2020 – Dec 2020
3DReid | Senior BIM Technician | Contract | Jun 2020 – Aug 2020
Robert Bird Group | Senior BIM Technician | Contract | Oct 2019 – Apr 2020
BG&E | Senior BIM Technician | Full-time | Feb 2019 – Oct 2019
Arcadis | BIM Technician | Full-time | Jan 2018 – Feb 2019
AECOM | BIM Technician | Full-time | Feb 2016 – Dec 2017

GRASSHOPPER STRATEGY

This Grasshopper Script is available upon request.

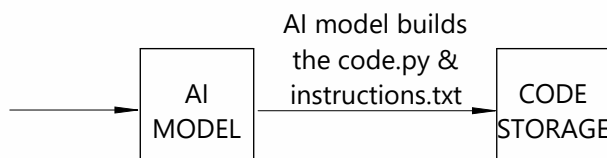


AI STRATEGY

AI Agent Step 1 - Build the Agent and its Toolkit

PROMPT

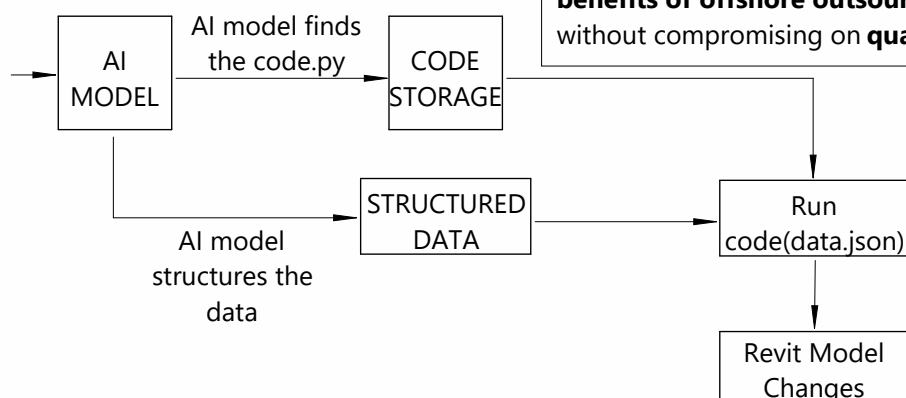
Modify Column Sizes.
Make me a code with in **Python** with **Revit API** that will find the columns by their unique mark and change their sizes.



AI Agent Step 2 - Run the Code

PROMPT

Modify Column Sizes.
The attached **.csv file/pdf/screenshot** contains the new set of column sizes.
Go to the toolkit folder and run the code that will modify them.



THE DETAILS

- This workflow uses AI Agents to directly make changes to our Revit model **from simple text and image inputs**.
- It cuts **hours** of time down to **minutes**.
- Regular and repetitive tasks such as column & beam sizing can be **offloaded from technicians** allowing them to focus on other details.
- When multiplied across a project, this workflow will **cut delivery times and total costs**.
- This method gives us the **cost benefits of offshore outsourcing** without compromising on **quality**.